



Organic Chemistry: Halogenoalkanes

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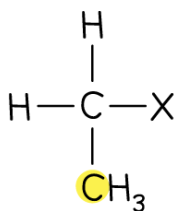
Halogenoalkanes – Introduction

Naming halogenoalkanes

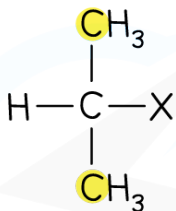
- We use the prefixes
 - **Fluoro** for fluorine
 - **Chloro** for chlorine
 - **Bromo** for bromine
 - **Iodo** for iodine
- The name of the halogenoalkane is based on the original alkane
 - So $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ would be 1-bromopropane
- The position of the halogen atom must be taken into account
 - So $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$ would be 2-bromopropane
- Finally the substituents are listed alphabetically
 - So $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{Br}$ would be 1-bromo-2-chloropropane

Classifying halogenoalkanes

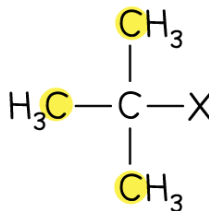
- Halogenoalkanes can be classified as **primary**, **secondary** or **tertiary** depending on the number of carbon atoms attached to the C-X functional group
 - A **primary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to one other alkyl group
 - A **secondary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to two other alkyl groups
 - A **tertiary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to three other alkyl groups



PRIMARY
HALOGENOALKANE



SECONDARY
HALOGENOALKANE



TERTIARY
HALOGENOALKANE

WHERE X IS A HALOGEN

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Examiner Tips and Tricks

You are also expected to be able to draw structural, displayed and skeletal formulae of halogenoalkanes



Your notes

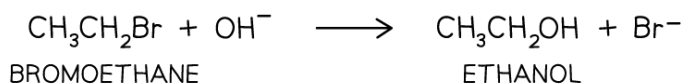


Halogenoalkanes – Reactions

- A **nucleophile** is an electron-rich species that can **donate** a pair of electrons
 - 'Nucleophile' means 'nucleus / positive charge loving' as nucleophiles are attracted to positively charged species
 - Nucleophilic** refers to reactions that involve a nucleophile

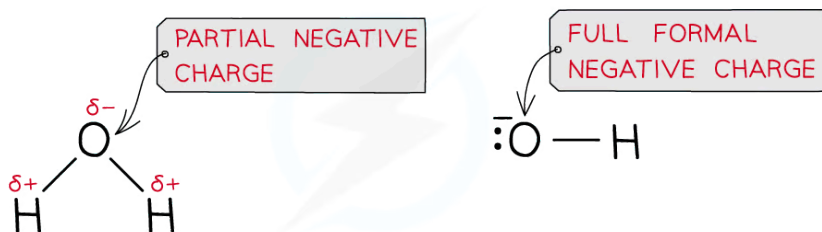
Formation of alcohols

- The nucleophile in this reaction is the **hydroxide ion**, OH^-
- An **aqueous** solution of sodium hydroxide (NaOH) or potassium hydroxide (KOH) with **ethanol** is used to form an alcohol
- This reaction is very slow at room temperature, so the reaction mixture is warmed
- This is an example of a **hydrolysis reaction** and the product is an alcohol
 - The rate of this reaction depends on the type of halogen in the halogenoalkane
 - The stronger the C-X bond, the slower the rate of the reaction
 - In terms of bond enthalpy, $\text{C-F} > \text{C-Cl} > \text{C-Br} > \text{C-I}$
 - Fluoroalkanes do not react at all, but iodoalkanes have a very fast rate of reaction



The halogen is replaced by the nucleophile, OH^-

- This reaction could also be done with water as the nucleophile, but it is very slow
 - The hydroxide ion is a better nucleophile than water as it carries a full negative charge
 - In water, the oxygen atom only carries a partial charge



A hydroxide ion is a better nucleophile as it has a full formal negative charge whereas the oxygen atom in water only carries a partial negative charge; this causes the nucleophilic substitution reaction with water to be much slower than the aqueous alkali



Your notes

Reaction with water

- The water molecule is a weak nucleophile, but it will eventually substitute for the halogen
- This occurs much more slowly compared to when warm aqueous sodium hydroxide is used
- An alcohol is produced
 - $RX + H_2O \rightarrow ROH + H^+ + X^-$
 - $CH_3CH_2Br + H_2O \rightarrow CH_3CH_2OH + H^+ + Br^-$
- If silver nitrate solution in ethanol is added to the solution, the silver ions will react with the halide ions as soon as they form, giving a silver halide precipitate
 - $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$

Formation of nitriles

- The nucleophile in this reaction is the **cyanide ion**, CN^-
- An **ethanolic solution** of **potassium cyanide** (KCN in ethanol) is **heated under reflux** with the halogenoalkane
- The product is a **nitrile**
 - E.g. bromoethane is heated under reflux with ethanolic potassium cyanide to form propanenitrile



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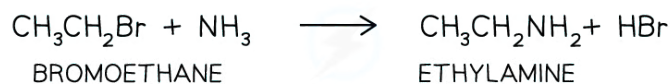
The halogen is replaced by a cyanide group, CN^-

- The nucleophilic substitution of halogenoalkanes with KCN adds an **extends the carbon chain** by adding an extra carbon atom
- This reaction can therefore be used by chemists to make a compound with one more carbon atom than the best available organic starting material

Formation of primary amines by reaction with ammonia

- The nucleophile in this reaction is the ammonia molecule, NH_3
- An **ethanolic solution** of **excess ammonia** (NH_3 in ethanol) is **heated under pressure** with a **primary** halogenoalkane
 - An excess of ammonia is used because the product is more reactive than ammonia so further substitution reactions could occur

- The product is a **primary amine**
 - E.g. bromoethane reacts with excess ethanolic ammonia when heated under pressure to form ethylamine

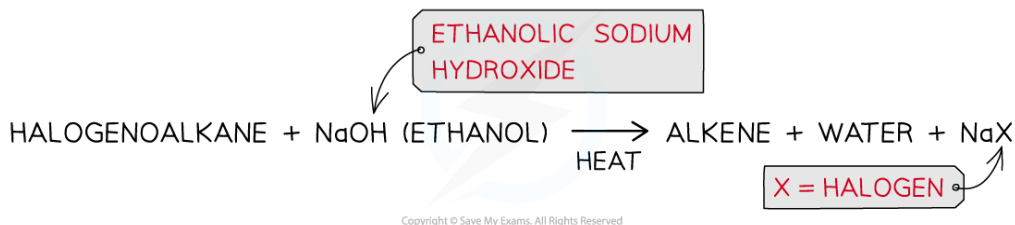


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The halogen is replaced by an amine group, NH_2

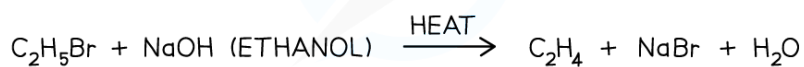
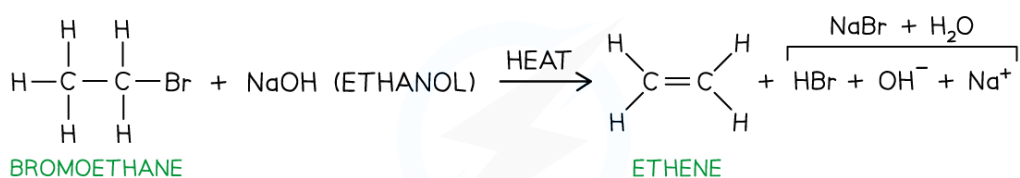
Formation of alkenes

- The halogenoalkanes are **heated** with **ethanolic sodium hydroxide** causing the C-X bond to break **heterolytically**, forming an X^- ion and leaving an alkene as an organic product
 - E.g. bromoethane is heated with ethanolic sodium hydroxide to form ethene



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Production of an alkene from a halogenoalkane by reacting it with ethanolic sodium hydroxide and heating it



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Hydrogen bromide is eliminated to form ethene



Your notes

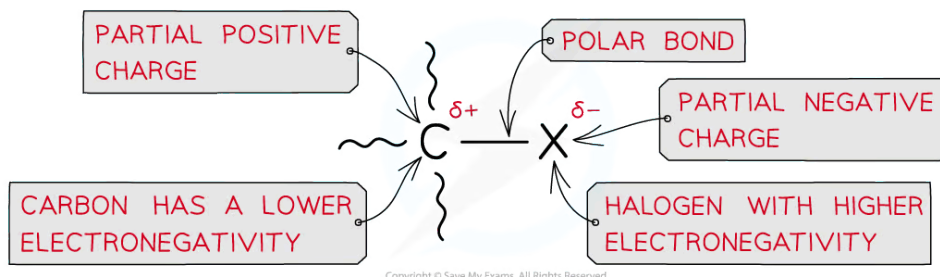
The Nucleophilic Substitution Mechanism



Your notes

Nucleophilic Substitution – Mechanism

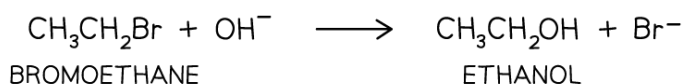
- A **nucleophilic substitution** reaction is one in which a **nucleophile** attacks a carbon atom which carries a **partial positive charge**
- An atom that has a **partial negative charge** is replaced by the nucleophile
- Halogenoalkanes will undergo nucleophilic substitution reactions due to the polar C-X bond (where X is a halogen)



Due to large differences in electronegativity between the carbon and halogen atom, the C-X bond is polar

Mechanism with aqueous potassium hydroxide

- In the following reaction a halogenoalkane reacts with aqueous alkali to form an alcohol



The halogen is replaced by a nucleophile, OH⁻

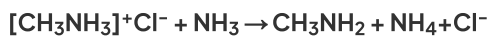
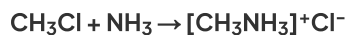
- The mechanism for the reaction is as follows



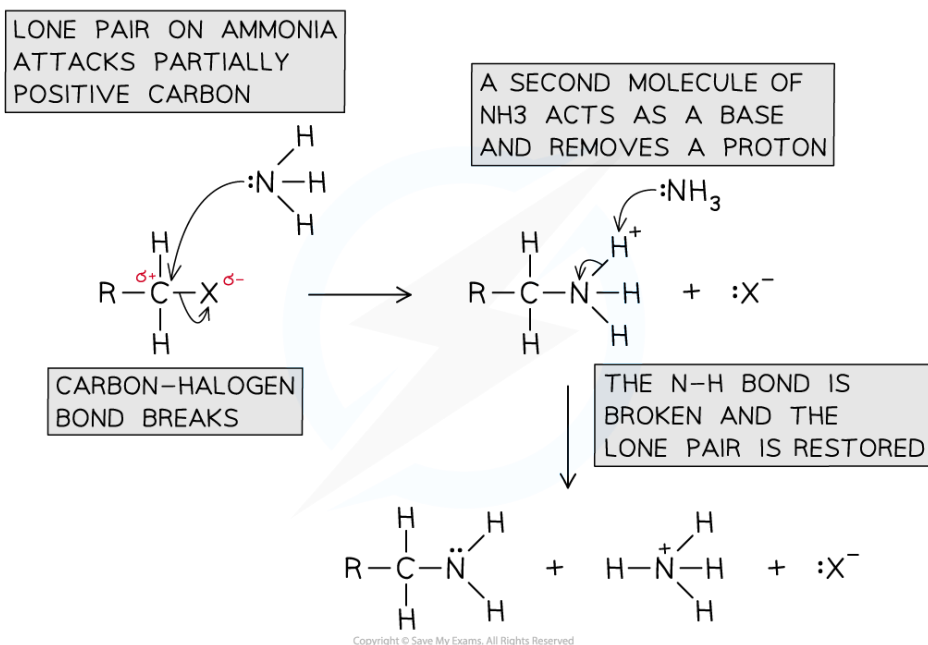
Nucleophilic substitution reaction of bromoethane and aqueous alkali (e.g. NaOH)

Mechanism with ammonia

- When ammonia reacts with a haloalkane a nucleophilic substitution reaction takes place forming a primary amine
 - For example chloromethane reacts with ammonia in two steps to make methylamine and ammonium chloride



- Excess** ammonia is used to prevent further substitution and favour the formation of a primary amine



The mechanism of nucleophilic substitution between ammonia and a halogenoalkane



Your notes



Halogenoalkanes – Hydrolysis

Measuring the rate of hydrolysis

- Acidified aqueous silver nitrate can be used to measure the rate of hydrolysis of halogenoalkanes
- The following method is used:
 - Set up three test tubes in a 50 °C water bath, with a mixture of ethanol and acidified silver nitrate
 - Add a few drops of a chloroalkane, bromoalkane and an iodoalkane to each test tube and start a stop watch
 - Time how long it takes for the precipitates to form
- Reacting halogenoalkanes with **aqueous silver nitrate solution** will result in the formation of a **precipitate**
- The **rate of formation** of these precipitates can also be used to determine the reactivity of the haloalkane

Haloalkane Precipitates Table

Halogenoalkane	Precipitate
Chlorides	White (silver chloride)
Bromides	Cream (silver bromide)
Iodides	Pale yellow (silver iodide)

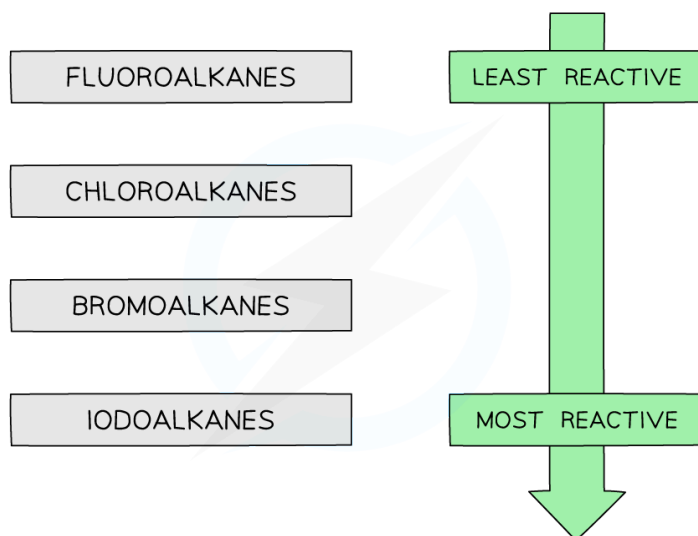
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- The precipitates will form as the reaction progresses and the halide ions are formed
 - The yellow silver iodide precipitate is the fastest **nucleophilic substitution** reaction
 - This is because the C-I bond has the lowest bond enthalpy, so it is the easiest to break and will cause the I⁻ ions to form the fastest
 - The white chloride precipitate is the slowest nucleophilic substitution reaction
 - This is because the C-Cl bond has the highest bond enthalpy, so it is the hardest to break and will cause the Cl⁻ ions to form the slowest
- Silver fluoride is soluble, so a precipitate will not be formed in this reaction

- This confirms that fluoroalkanes are the least reactive and iodoalkanes are the most reactive halogenoalkanes
- It can be predicted that the formation of silver astatide would be even quicker than silver iodide



Your notes



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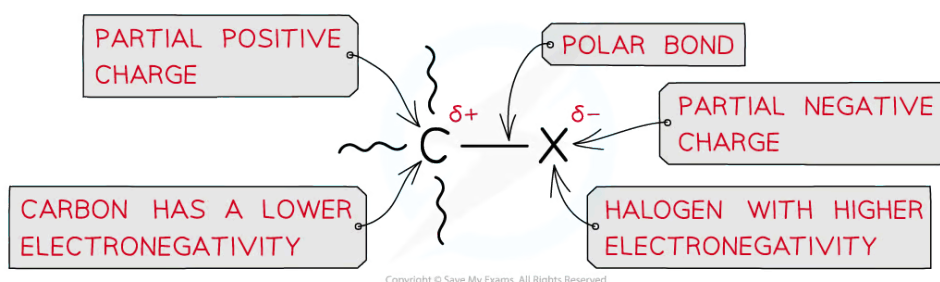
The trend in reactivity of haloalkanes

- A similar experiment can be set up to compare the rate of hydrolysis in primary, secondary and tertiary halogenoalkanes
 - The halogen involved and the overall molecular formula of the halogenoalkane must remain the same
 - E.g. 1-chlorobutane (primary), 2-chlorobutane (secondary) and 2-chloro-2-methylpropane (tertiary)
- Results would show that:
 - The tertiary halogenoalkane / 2-chloro-2-methylpropane reacts the quickest
 - The primary halogenoalkane / 1-chlorobutane reacts the slowest
 - Therefore, the secondary halogenoalkane / 2-chlorobutane is in the middle
 - Although, the explanation for this is not required at AS level, it is required at A-level
 - The rate of hydrolysis is determined by the use of either an S_N1 or an S_N2 reaction mechanism



Reactivity & Bond Enthalpy

- Halogenoalkanes contain at least one halogen atom covalently bonded to a carbon atom within the molecules structure
- Due to the difference in electronegativity between the halogen and carbon atoms:
 - The carbon-halogen, C-X, bond is polar
 - The carbon atom has a $\delta+$ charge
 - The halogen atom has a $\delta-$ charge



Due to large differences in electronegativity between the carbon and halogen atom, the C-X bond is polar

- As the electronegativity down the group decreases, so does the polarity of the bond
 - This suggests that fluoroalkanes would be the quickest to react but the strength of the bond is another important and greater factor

Bond Enthalpy

- The halogenoalkanes have different **rates** of **substitution reactions**
- Since substitution reactions involve **breaking** the **carbon-halogen** bond the **bond energies** can be used to explain their different reactivities

Halogenoalkane Bond Energy

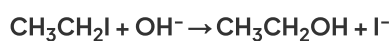


Your notes

Bond	Bond Energy (kJ mol ⁻¹)
C-F	467 (strongest bond)
C-Cl	346
C-Br	290
C-I	228 (weakest bond)

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- The table above shows that the C-I bond requires the least energy to break, and is therefore the **weakest** carbon-halogen bond
- During substitution reactions, the C-I bond of the halogenoalkane will therefore **heterolytically** break to form the alcohol
 - For example



- The C-F bond, on the other hand, requires the most energy to break and is, therefore, the **strongest** carbon-halogen bond
- Fluoroalkanes will, therefore, be less likely to undergo substitution reactions